MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 6236

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Reflex, 100 mW rated power output, external cavity

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Ec3	Er	Ib	Ic1	T(g3)
Unit:	V	Vdc	Vdc	Vdc	Vdc	mAdc	mAdc	°C
Maximum:	6.8	25	1,100	1,100	-600	22	6.5	160
Minimum:	5.7	---	---	---	-15	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1

TEST CONDITIONS: See note 4

Parameter:	Ef	Ec1	Ec2	Ec3	Er	Ib	F
Unit:	V	Vdc	Vdc	Vdc	Vdc	mAdc	MHz
Test condition 1:							
Maximum:	---	See	---	---	-145	---	4,390
:	6.3	note	1,000	1,000	---	20	---
Minimum:	---	3	---	---	-60	---	4,190
Test condition 2:							
Maximum:	---	See	---	---	-520	---	7,685
:	6.3	note	1,000	1,000	---	20	---
Minimum:	---	3	---	---	-350	---	7,315

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
4280	Electronic tuning range	---	1, 2	Er = 50% max Po	ΔF	10	---	MHz
4027	Temperature coefficient	5	---		ΔF/ΔT	---	±0.1	MHz/°C
	Quality conformance inspection, part 1	1	---					
1256	Electrode current (positive grid)	---	1, 2		Ic1	---	6	mAdc
1261	Electrode voltage (grid)	---	1, 2		Ec1	4	18	Vdc
4213	Reflector voltage	6, 7 6, 8	1 2		Er Er	-60 -350	-145 -520	Vdc Vdc
4229	Total reflector current	9	---		Ir	---	7	μAdc
4250	Power output	6, 7 6, 8	1 2		Po Po	90 100	--- ---	mW mW
4214	Cathode emission	---	1	Ef = 5.8 V	ΔIb/Ib	---	15	%
	Quality conformance inspection, part 2							
1031	High-frequency vibration	10	---	No voltage	---	---	---	---
1261	Electrode voltage (control grid at Ib cutoff)	---	---	Ef = 6.3 V; Ec2 = Ec3 = 1,000 Vdc; Ib = 1 mAdc (max)	Ec1	---	-23	Vdc
1301	Heater current	---	---	Ef = 6.3 V	If	520	640	mAdc
1336	Heater-cathode leakage	---	---	Ehk = ±45 Vdc	Ihk	---	100	μAdc
---	Pulsed operation	11	---	F = 3,800 MHz F = 7,600 MHz	Er Er	-50 -345	-135 -510	Vdc Vdc
	Quality conformance inspection, part 3							
---	Life test	2	1	Group C	t	500	---	hrs
---	Life-test end point:							
4250	Power output	6, 8	2		Po	75	---	mW

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NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.
2. Total current to the interaction gap electrodes (Ib) shall be adjusted to 20 mA by adjustment of Ec1 at the beginning of life test. Ec1 shall be changed during life test to maintain total current (Ib) of 20 mA to the interaction gap electrodes. The maximum bias of 18 volts on grid 1 shall be observed.
3. Ec1 shall be adjusted for a total current (Ib) of 20 mA to the interaction gap electrodes with the tube in oscillation. Er shall be adjusted for maximum power output.
4. The tube shall be oscillating in the 2-3/4 cycles reflector transit mode.
5. With test cavity as shown on Drawing 177-JAN, the temperature coefficient of the cavity shall be -11 kHz per °C maximum. The temperature coefficient of the tube shall be considered to be the difference between the characteristics of the cavity alone and the tube with the cavity.
6. Er shall be adjusted for maximum power output into a fixed 50-ohm load with VSWR$\leq$1.25.
7. With test cavity as shown on Drawing 176-JAN, cavity 2B, the loop and plug shall be adjusted for maximum power output into a fixed 50-ohm load with VSWR$\leq$1.25.
8. With test cavity as shown on Drawing 176-JAN, cavity 4, the loop and plug shall be adjusted for maximum power output into a fixed 50-ohm load with VSWR$\leq$1.25.
9. This shall be a nonoscillating test. The tube shall be tested as specified in note 7 at a reflector voltage, Er, greater than -60 volts.
10. After performance of the specified test, the tube shall satisfy the requirements of quality conformance inspection, part 1, as specified herein.
11. The tube shall be oscillating in the 2-3/4 cycles reflector transit mode during this test, and the top of the pulse envelope as viewed on a synchroscope shall be essentially flat. The jitter or multiple shift in the rise time shall be less than 0.3 μs at any particular frequency throughout the frequency range specified herein, and there shall be no evidence of loss of pulse or intermittent pulsing. The pulse amplitude shall not very more than 2 dB from the CW amplitude as viewed on a synchroscope. The variation in pulsed output starting time for any particular tube, as the frequency is varied throughout the range specified herein, shall not exceed 0.5 μs.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army -
Air Force - 11
DSA - ES

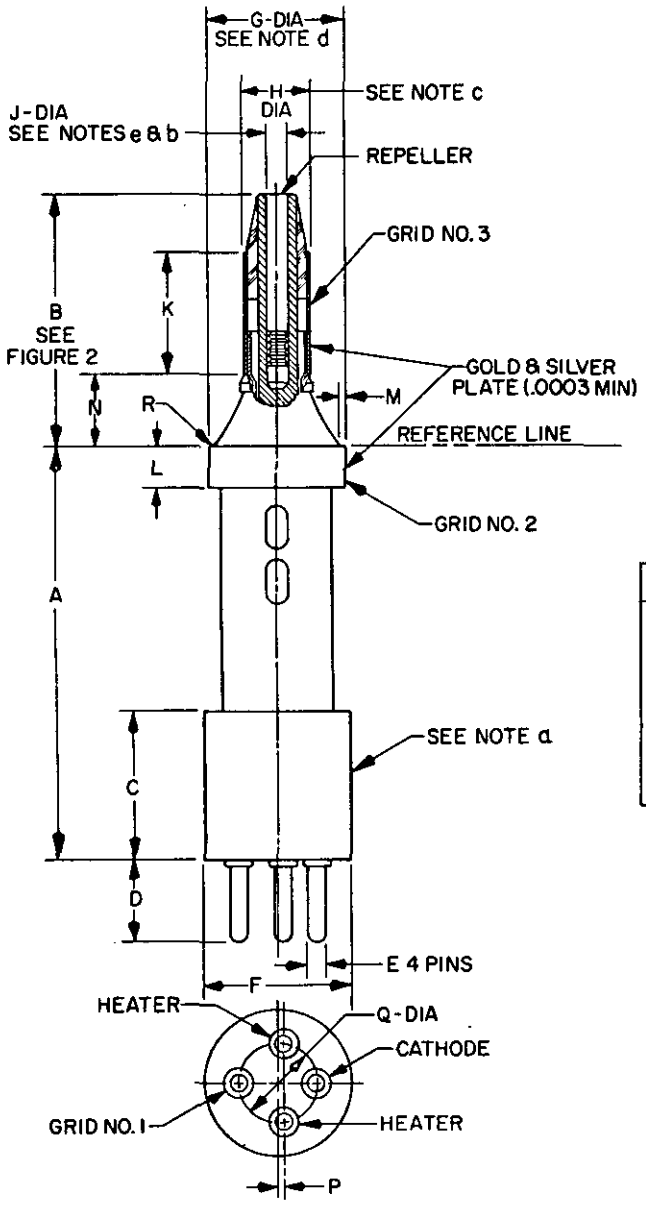
User activities:
Army - WC
Navy - AS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

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Electrical connections	
Pin No.	Element
1	g
2	h
3	k
4	h
Lower ring	g2
Upper ring	g3
Jack	r

FIGURE 1. Outline drawing of electron tube type 6236.

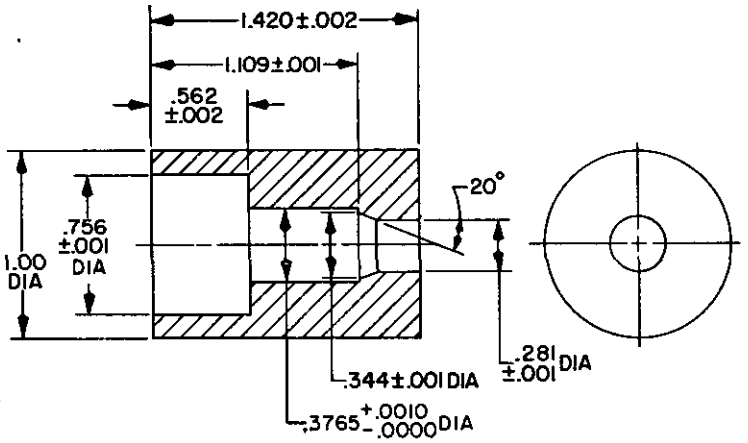
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
C	.720 (18.29)	.780 (19.81)
D	.467(11.86)Nom.	
E	.090 (2.29)	.097 (2.46)
F	.782 (19.86)	.844 (21.44)
P	.030 (.76)Nom.	
Q	.415(10.54)Nom.	
Quality conformance inspection, part 1 (see note 1)		
H	.372 (9.45)	.376 (9.55)
M	.020 (.51)	
N	.340 (8.64)	.407 (10.34)
R	.003 (.08)	.008 (.20)
Quality conformance inspection, part 2		
A	1.860 (47.24)	2.360 (59.94)
B	1.360 (34.54)	1.420 (36.07)
G	.782 (19.86)	.786 (19.96)
J	.105 (2.67)	.113 (2.87)
K	.575 (14.61)	.675 (17.15)
L	.190 (4.83)	.230 (5.84)

NOTES:
(a) Low-loss phenolic base.
(b) This diameter applies for a depth of .312(7.92 mm) min.
(c) Reference diameter.
(d) This diameter to be concentric to H within .005(.13 mm).
(e) This diameter to be concentric to H within .015(.38 mm).

FIGURE 1. Outline drawing of electron tube type 6236 - Continued.

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NOTES:
(a) The internal diameters shall be concentric within .0005(.01 mm) TIR.
(b) Tube shall enter far enough to seat on the leading edge of the G2 ring.

FIGURE 2. Reflector insulator gage.

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